

Predicting Hourly Bike Sharing Demand in Seoul Using Machine Learning

LILY BRYAN

CISC 5800

Fordham University

Machine Learning

INTRODUCTION

Problem & Motivation

The Main Goal

- Seoul's bike sharing system is one of the largest in Asia
 - 20,000+ bikes
 - 1,500+ stations
 - Operates 24/7
- Inaccurate demand prediction can lead to wasted resources
 - Too many bikes in areas with low demand & vice versa
- Accurately predicting bike demand is essential for efficient bike allocation



Dataset

Seoul Bike Sharing Dataset - UCI Machine Learning Repository

Key Facts

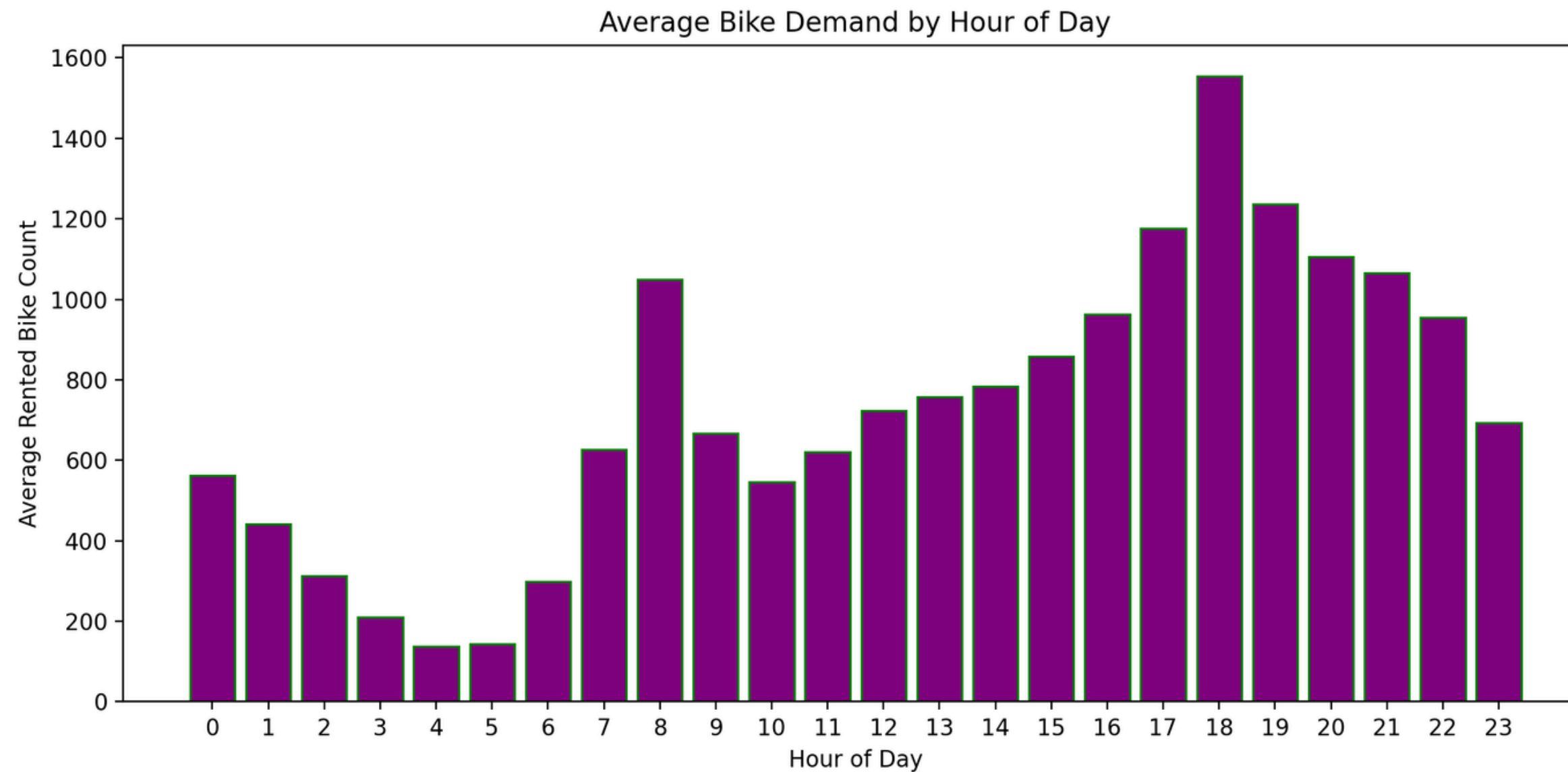
- 8,760 hourly instances
- 13 features, 1 target variable (Rented Bike Count)
- December 2017 - November 2018
- No missing values

Feature Types

- 10 numerical features:
 - Temperature
 - Humidity
 - Wind Speed
 - Visibility
 - Dew Point
 - Solar Radiation
 - Rainfall
 - Snowfall
 - Hour
 - Date
- 3 categorical features:
 - Season
 - Holiday
 - Functioning Day
- Target:
 - Rented Bike Count (0 to 3,556 bikes / hour)

Exploratory Data Analysis

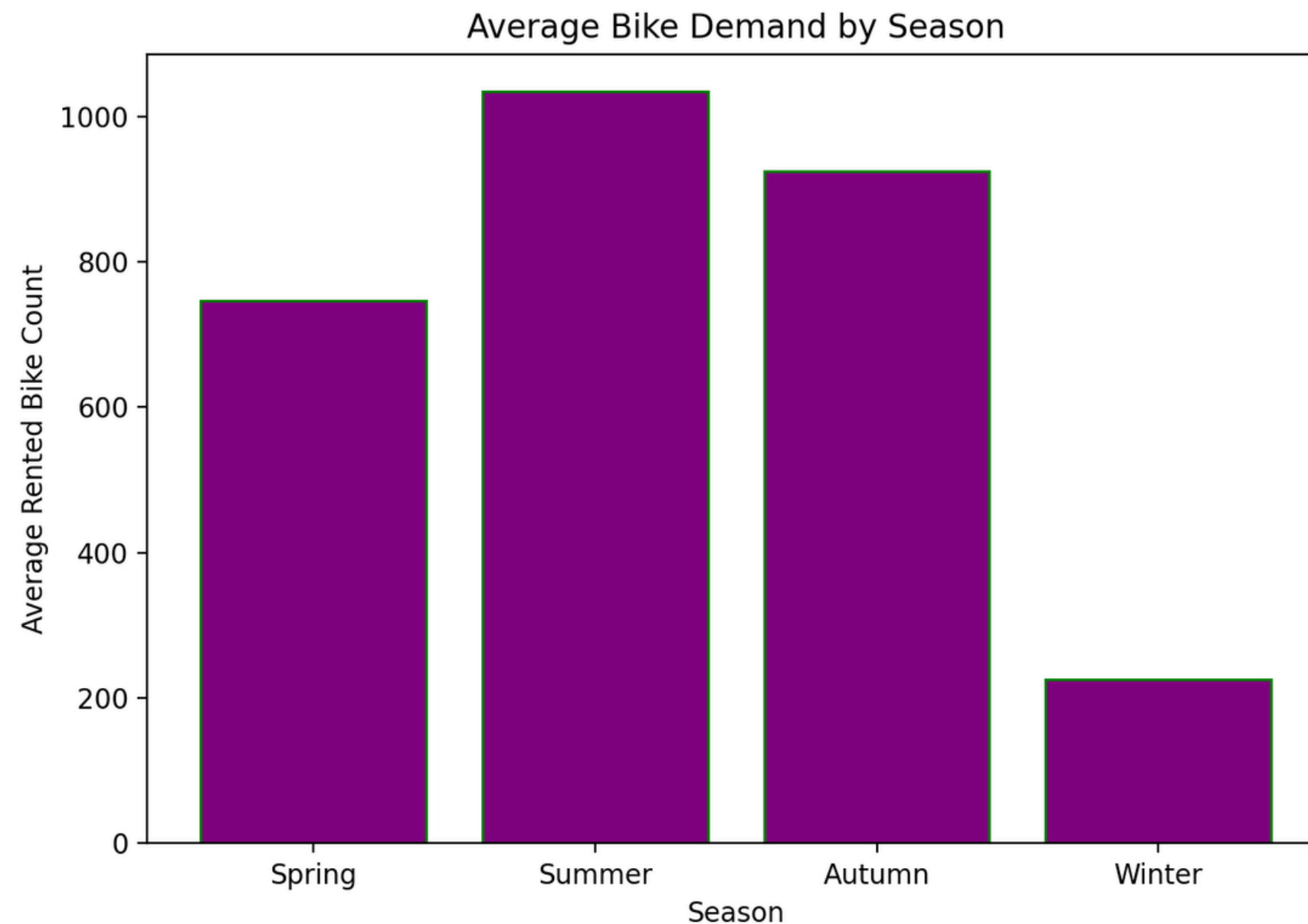
Demand by Hour



- Clear peaks around 8:00AM (~1,050 bikes) and 6:00PM (~1,550 bikes)
 - Commuter rush hours
- Lowest in early morning hours (around 3:00AM to 5:00AM)
- Bikes are clearly heavily used by commuters
- Nonlinear relationship with demand

Exploratory Data Analysis

Demand by Season



- Summer has highest demand (~1,035 bikes / hour)
- Winter has lowest demand (~225 bikes / hour)
 - Winter demand is 5 times lower than Summer demand
- Temperature is strongest individual predictor of demand
 - $r = 0.56$
- However, no single feature fully explains demand

Preprocessing

Nonfunctioning Days

- Removed 295 non-function day rows
- System downtime, not real demand

One-Hot Encoding

- Season and Holiday
- From categorical to numerical

Time Ordered Train / Test Split

- 70 / 15 / 15
- Done in chronological order to prevent data leakage

Feature Scaling

- Used StandardScaler on training data only, applied to validation and test data

Linear Regression & Lasso Regularization

Linear Regression

- Baseline Ordinary Least Squares model:
 - **MSE = 285,321**
 - **$r^2 = 0.34$**
- Temperature is the strongest predictor
- Counterintuitive coefficient findings: Snowfall (+) and Solar Radiation (-)
- Winter Season and Temperature have r value of -0.74, confirming multicollinearity

Lasso Regularization

- Grid search over alpha values
 - Best alpha: 0.1
- Model results:
 - **MSE = 275,230**
 - **$r^2 = 0.3663$**
 - Slight improvement over baseline
- Only Spring feature eliminated
 - Almost all features carry meaningful information

Logistic Regression

Class	Precision	Recall	F1
Low Demand	0.66	0.67	0.67
High Demand	0.87	0.87	0.87
Accuracy	0.81		

- Threshold = 542 (median), High vs Low demand
- Class imbalance in validation set
 - 912 high demand vs. 358 low demand instances
- Low demand is harder to classify
 - Occurs across all seasons and weather conditions

Kernel Ridge Regression

	Validation
MSE	158,281
r^2	0.6356

- Kernel trick:
 - Maps features into higher dimensional space without manually creating interaction features
- Radial Basis Function (RBF) Kernel chosen
 - Measures similarity between data points using Euclidean distance
 - Works well for smooth, continuous patterns like weather and demand
- Grid search over alpha (0.001 to 100) and gamma (0.0001 to 1)
 - Best alpha: 0.1
 - Best gamma: 0.1
- r^2 value almost doubled from 0.34, confirming nonlinear relationship

Results & Comparison

Model	Set	MSE	r^2
Linear Regression (OLS)	Validation	285,321	0.34
Lasso ($\alpha = 0.1$)	Validation	275,230	0.37
KRR ($\alpha = 0.1, \gamma = 0.1$)	Validation	158,281	0.64
KRR ($\alpha = 0.1, \gamma = 0.1$)	Test	166,003	0.46

- Test set R^2 drop may be attributed to late Autumn seasonal shift, as the model is trained on Spring/Summer/early Autumn data

Conclusions & Future Work

Key Findings

- KRR with RBF kernel was the best performing model ($r^2 = 0.64$ on validation set)
- Nonlinearity confirmed, linear models insufficient for this problem

Limitations

- Only one year of data
 - Model barely sees late Autumn patterns
- Seasonal distribution shift
- Outside factors not captured
 - events, construction, transit disruptions, etc

Future Work

- Multiple years of data to capture seasonal cycles
- Add public transit and event data as features
- Neural network for more complex interactions
- Station-level prediction for targeted bike rebalancing